Credit Structuring Methods in Prime Non-Agency MBS

February 23, 2007

Sharad Chaudhary
212.847.5793
sharad.chaudhary@bankofamerica.com

RMBS Trading Desk Strategy

Ohmsatya Ravi
212.933.2006
ohmsatya.p.ravi@bankofamerica.com

Qumber Hassan
212.933.3308
qumber.hassan@bankofamerica.com

Vipul Jain
212.933.3309
vipul.p.jain@bankofamerica.com

Sunil Yadav
212.847.6817
sunil.s.yadav@bankofamerica.com

Ankur Mehta
212.933.2950
ankur.mehta@bankofamerica.com

RMBS Trading Desk Modeling

ChunNip Lee
212.583.8040
chunnip.lee@bankofamerica.com

Marat Rvachev
212.847.6632
marat.rvachev@bankofamerica.com

- Credit enhancement in non-agency mortgage securitizations has come from a number of sources in the past but the provision of internal credit enhancement through the creation of subordinate classes has become the dominant way to shelter investors from credit risk in such transactions.
- The modern day tool-kit of structuring techniques to distribute credit risk in a typical non-agency structure starts with the creation of senior and subordinate tranches. Any losses realized on the underlying collateral are first used to write down subordinate tranches. Credit support for senior tranches is further augmented by locking out prepayments or all principal payments to the subordinate tranches for a number of years, or until certain conditions are met. These payments to the subordinate tranches are further regulated by **stepdown** and **trigger** conditions.
- The two credit risk distribution structures that dominate the market are the **shifting interest** structure and the **over-collateralization** structure. These structures represent variations on a basic senior/subordinate structure and are distinguished (in part) by how subordinate tranches are paid. In a **shifting interest** structure, the pro rata share of prepayments due to the subordinate tranches is shifted to the senior tranches for a period of time to accelerate the amortization of these classes. In an **over-collateralization (OC)** structure, subordinate classes start receiving principal paydowns such that all bonds maintain a specified multiple of the original credit enhancement.
- The primer discusses how these structuring techniques are used to create different types of credit structures. We start with a high-level conceptual overview of how the shifting interest and over-collateralization structures distribute credit risk over different tranches in a deal. This is followed by a detailed analysis of how the cash flows to comparable tranches in the two structures are affected by different prepayment and loss scenarios. A final section looks at how the underlying collateral (fixed-rate or adjustable-rate mortgages) affects the cash flow profiles of various tranches in a shifting interest structure.
- The subordinate tranches of an OC structure will typically have shorter weighted-average lives and worse negative convexity than similarly rated tranches of a shifting interest structure. On the other hand, due to its reliance on an IO-like cash flow for providing additional credit enhancement, the OC structure provides better protection in slow prepayment scenarios for both senior and subordinate tranches relative to the shifting interest structure.

I. INTRODUCTION

Historically, credit enhancement in non-agency mortgage securitizations came from a number of different sources including corporate guarantees, letters of credit, surety bonds, pool insurance, spread accounts, reserve funds and subordinate classes.[1] Over time, the provision of internal credit enhancement through the creation of subordinate classes has become the dominant way to shelter investors from the credit risk in a typical non-agency mortgage-backed securities (MBS) transaction. In a nutshell, the subordinate class provides internal credit enhancement to senior classes in the structure by getting first priority on any losses experienced by the underlying collateral. Conceptually, this technique of redistributing losses from the underlying collateral to different tranches of the deal such that some tranches carry less default risk than others is similar to the methods employed in agency CMO structures to redistribute prepayment risk. However, this reparcelling of losses is far from the only method used to distribute credit risk in a typical non-agency structure. The modern day laundry list of techniques for achieving this goal can be summarized as follows:

- **Senior/Subordinate ("Senior/Sub") Structures**. The most basic way of providing credit enhancement through the cash flow structuring process ("internal credit enhancement") is to create senior and subordinate tranches from a pool of mortgages with realized losses being allocated to the most subordinate tranche of the deal outstanding at that particular juncture. In addition, interest payments to the senior tranches have priority over interest payments to the subordinate tranches. A similar statement holds for principal payments. As a result of these two factors, senior tranches have less exposure to credit risk and consequently receive higher credit ratings than subordinate tranches.
- **Lockout of Principal Payments to Subordinates**. In modern structures, credit support for senior tranches is further augmented by locking out unscheduled principal (prepayments) or all principal payments to the subordinate tranches for a certain number of years, or until certain conditions are met. A lockout mechanism effectively enhances the subordination level of the senior tranches as the underlying collateral is paid down. In addition, lockout provides a defense against adverse selection from the credit perspective as borrowers with better credit may prepay out of the pool quicker than borrowers with poor credit. In some instances, lockout also provides protection against back-loaded losses. For example, the low mortgage payments required of interest-only (IO) and Option ARM borrowers in the initial years may mitigate credit losses initially. Once the lockout period is over, or the conditions that override the lockout period have been met, subordinates can start receiving the payments that they have been locked out of.[2] This is called a **stepdown**.

[1] Subordinate classes were first introduced in 1986.
[2] Depending upon the lockout conditions, this payment can consist of just prepayments, or of all scheduled and unscheduled principal.

- **Payment Rules for Subordinates.** There are a couple of different variations to the vanilla senior/subordinate structure that influence how subordinates receive principal once stepdown occurs. In a **shifting interest** structure, the pro rata share of prepayments due to the subordinate tranches is shifted to the senior tranches for a period of time in order to accelerate the amortization of these classes. In an **over-collateralization** structure, subordinate classes start receiving principal paydowns such that all bonds maintain a specified multiple (typically 2x) of the original credit enhancement. On any payment date, the distribution of cash flows to senior and subordinate tranches is arranged to be such that the maximum proportion of bonds achieve or restore this specified multiple of the original credit enhancement. The distribution of cash flows that achieves this target is referred to as the **optimal distribution**.

- **Over-collateralization.** Over-collateralization ("OC") structures were briefly discussed in the previous paragraph and are a popular alternative to the shifting interest senior/subordinate structure. We start with a senior/subordinate structure but in addition to having subordinate classes, also provide credit enhancement through two other means: over-collateralization (creating fewer bonds than the available balance of the collateral) and payments from IO strip that consists of the difference between the weighted-average coupon (WAC) on the underlying pool of mortgages and the WAC of the bonds in the deal. This IO strip is known as **excess spread**.

- **Triggers:** Triggers are checks on the delinquency rate and cumulative losses in a pool of mortgages in order to ensure that senior tranches remain protected in the event of a credit meltdown. Triggers protect senior tranches by preventing or suspending stepdown when they are tripped.

In this primer, we discuss how the above techniques are applied to create different types of credit structures. The primer begins with an overview of the senior/subordinate concept and then reviews shifting interest and over-collateralization structures. The emphasis is on providing a high-level conceptual understanding of how these structures redistribute credit risk over the different tranches in a deal. To obtain a more refined understanding of how these structures work in practice, the next section compares cash flows on comparable tranches in the two structures under different prepayment and loss scenarios. A final section analyzes the effect of collateral type on the cash flow profiles of the various tranches in a deal by comparing structures backed by fixed-rate and adjustable-rate collateral respectively. The document concludes with two appendices. The first appendix describes how cash flows are structured at the loan level to create the structures described above while the second details the payment rules used in our analysis of shifting interest and OC structures.

II. CREDIT STRUCTURING METHODS

The collateral backing Non-Agency CMOs consists of prime or near-prime credit loans that are not securitized in GSEs pools because their loan sizes exceed the conforming limit ("Jumbo" loans), their collateral characteristics lie outside the periphery of GSE underwriting guidelines ("alt-A" loans), or because better execution is available in the Non-Agency market. In the absence of a GSE credit guarantee, some form of credit enhancement must be provided when these loans are securitized. The standard technique of providing credit enhancement in Non-Agency CMOs through the structuring process[3] is by creating a "Senior/Sub Shifting Interest" structure, although "Over-Collateralization" structures have also gained popularity in the past few years. The goal of this section is to provide a conceptual overview of these two structuring techniques. The next section delves into more of the nitty-gritty details by conducting a comparative cash flow analysis of the two different structures.

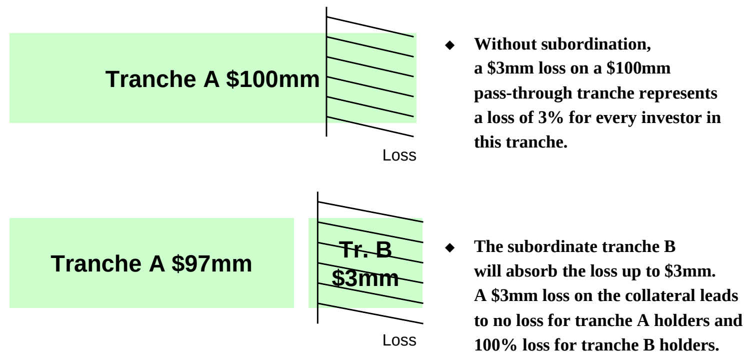
Fundamental Building Blocks: Senior and Subordinate Classes

The basic building block of a Non-Agency CMO structure is the creation of senior and subordinate tranches. In a Senior/Sub structure, the subordinate tranches ("Subs") protect the senior tranches ("Seniors") by absorbing all default-related losses until their principal is exhausted. The size of the subordinate tranches is determined by the rating agencies and currently constitutes between 3-4% of the deal principal for prime 30-year jumbo fixed-rate loans. For example, a $100 million loan package with 3% subordination would have a total credit enhancement of $3 million (= 100 x 0.03) supporting a senior tranche balance of $97 million (= 100 x 0.97). All losses up to $3 million would be absorbed by the subordinate tranches. For losses greater than $3 million, the senior tranche absorbs the difference between the total loss and $3 million. As a result of this credit protection, the senior tranche is rated AAA. Figure 1 provides a visual illustration of this technique for credit enhancement.

[3] As opposed to external means like pool insurance or a spread account.

Figure 1: Credit Enhancement through Tranche Subordination

- Senior tranches are protected from credit losses through subordinated tranches
  - Subordinate tranches get all the losses, until they have no outstanding principal
  - Senior tranches have less exposure to losses, while subordinated tranches have magnified exposure with respect to losses



Source: Banc of America Securities

In practice, the tranching shown in Figure 1 represents a very simplified version of what happens in reality with both the senior and subordinate classes of the structure subject to further structuring depending upon market demand.[4] For example, in the senior/subordinate shifting interest structure that we will shortly discuss in detail, the subordinate is further tranched into a "6 pack" – six tranches with ratings of AA, A, BBB, BB, B, and NR. The most subordinate bond ("NR") is called the **first loss** tranche and is not rated by the rating agencies. Figure 2 summarizes current subordination levels for the different tranches of a recently issued Jumbo A 30-year and 15-year deal.[5] So, for example, the second column of the figure shows that the NR bond is unprotected and immediately absorbs all losses, while the AA tranche only takes a loss if 1.35% of the deal principal has already been used to cover losses.

[4] Some of the details associated with this structuring process can be found in Appendix A.
[5] These subordination levels are determined by the Rating Agencies and are a function of both the macroeconomic environment and the underlying collateral.

Figure 2: Recent Tier 1 Subordination Levels for a Simple Senior/Sub Structure

Credit Rating	30-year Tier 1 Jumbo	15-year Tier 1 Jumbo
AAA	3.40%	1.85%
AA	1.35%	0.70%
A	0.80%	0.50%
BBB	0.50%	0.30%
BB	0.30%	0.20%
B	0.15%	0.10%
NR	0.00%	0.00%

Source: Banc of America Securities

Senior /Subordinate Shifting Interest Structure

In the most common enhancement of the basic senior/subordinate structure, seniors are further protected by a **shifting of prepayments** (also called a **shifting interest** structure). In this structure, the pro rata share of prepayments due to the subordinate tranches is shifted to the senior tranches for a period of time in order to accelerate the amortization of these classes and increase the amount of credit support available to them.[6] Expressed differently, the subordinates are **locked out** of receiving unscheduled principal payments (prepayments) for a period ranging from 5 to 10 years depending upon whether the underlying collateral is fixed-rate mortgages (FRMs) or adjustable-rate mortgages (ARMs). The timing of the lockout period is intended to mirror the seasoning ramp of losses and delinquencies that we typically see on a mortgage pool and make sure that the senior tranches receive protection during a critical early period until losses stabilize. A shifting interest structure requires less upfront subordination and thus makes the deal more economically efficient.

A typical shifting interest schedule based on a 10-year and 5-year lockout period is shown in Figure 3 below. In the 5-year lockout case, the senior tranche receives 100% of the subordinates' pro rata share of prepayments for the first 5 years. As the figure shows, this percentage steadily decreases until subordinates become entitled to 100% of their pro rata share of prepayments after 10 years. In case of the 10-year lockout (the second column in Figure 3), the senior tranche receives 100% of the subordinates' pro rata share of prepayments for the first 10 years and the percentage is then gradually reduced using the schedule listed in the figure.

[6] Because their relative proportion in the deal decreases.

Figure 3: Typical Shifting Interest Prepayment Lockout Schedules

5-year Lockout	10-year Lockout	Shift %
Month <=	*Month <=*	
60	120	100
72	132	70
84	144	60
96	156	40
108	168	20
120	180	0

Source: Banc of America Securities

In addition to the lockout mechanism specified above, a senior/subordinate shifting interest structure will also often come equipped with two **triggers** that act as a safety valve for the senior tranche. The typical trigger tests that are used are as follows:

- **Delinquency Test:** 6-month average 60+ Day Delinquent Balance less than 50% of Current Subordinate Balance.
- **Cumulative Loss Test:** If the deal has a 5-year lockout period, losses do not exceed a specified percentage of the initial subordinate balance based on the following caps: 30% in the first 6 years, 35% at the end of 7 years, 40% at the end of 8 years, 45% at the end of 9 years and 50% at the end of 10 years.

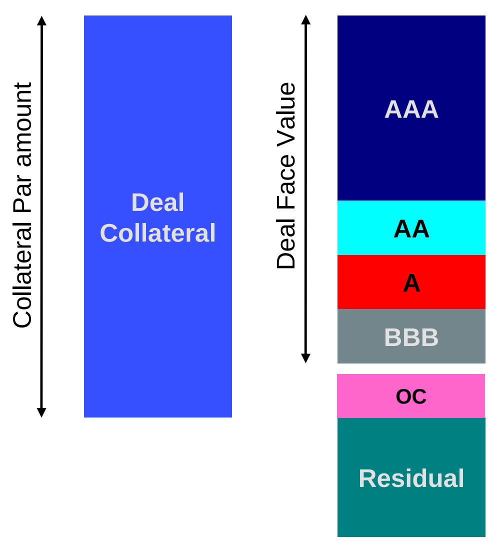
Both triggers are evaluated at every payment date and both have to be passed for the stepdown schedule described in Figure 3 to occur. Concretely, if a trigger is tripped, the subordinate tranches stop receiving prepayments (assuming the initial lockout period has ended) and these payments are funneled to the seniors. Prepayments to the subordinates that conform to the lockout schedule are only resumed once both the trigger tests are passed.

Although the senior/subordinate shifting interest structures used for securitizing non-agency fixed-rate and ARM collateral share several features, they usually differ from each other in terms of how the lockout period is specified. The lockout period for subordinates in deals backed by fixed-rate collateral is typically 5 years as opposed to 7-10 years in the case of deals backed by ARM collateral. More importantly, the 5 year lockout period for fixed-rate collateral represents an *absolute* lockout period i.e., early release of unscheduled principal payments to the subordinates is prohibited until the senior tranches are completely paid down. Structures backed by ARM collateral conform to a different convention. Here, release of prepayments to the subordinates can take place earlier than the end of lockout period if some previously specified enhanced levels of subordination are met and triggers are passed. Typically, if subordination levels reach two times (2x) the initial credit support during the first 3 years, then subordinates receive 50% of their pro rata share of prepayments. After the first three years, reaching 2x of the original subordination levels results in the subordinates receiving 100% of their pro rata share of prepayments. If subordination levels fall below 2x, we revert to the original 7-10 year lockout period.

Over-Collateralization Structures

While over-collateralization (OC) structures have been frequently used to securitize subprime mortgages, they have only recently become a popular credit structuring alternative for deals backed by prime mortgages. As illustrated in Figure 4, similar to a shifting interest structure, an OC structure starts out by splitting the collateral into senior and subordinate tranches with a decreasing degree of subordination as we move down in credit rating from the senior most to the junior most tranche. One big difference between the OC and the classic shifting interest structure lies in the existence of an OC account. The over-collateralization account is the difference between the principal balance on the collateral and the principal balance on the deal. Think of the OC amount as a "slush" fund or a reserve account which can be used to provide credit enhancement to the deal. Another difference is the presence of a **residual (first loss)** tranche in the OC structure. The residual tranche does not have a principal balance or a fixed coupon to start with and is backed by all the cash flows that remain after all the principal and interest payments due to the trust have been satisfied.[7]

Figure 4: High-level Overview of an OC Structure



Source: Banc of America Securities

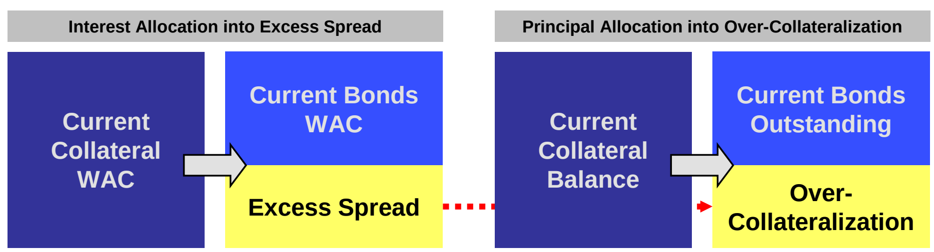
The cash flow mechanics of how credit enhancement is provided in the OC structure are somewhat more involved than in the shifting interest case. In addition to the subordinate bonds, the OC account and **excess spread (excess interest)** are used to provide credit enhancement to all the bonds in a deal. Figure 5 provides a somewhat simplified illustration of how over-collateralization and excess spread are created in the structuring process. **Excess spread** (**excess interest)** is an IO cash flow with a coupon equal to the

[7] In Figure 4 and the discussion we separate the OC account and Residual as if they were separate. This is more for illustration purpose since the residual holder has ultimate claim over OC to the extent that it remains after paying off credit losses on senior tranches. In other words, the residual and OC should be considered as one class.

difference between the weighted-average coupon on the collateral and the weighted-average coupon on the outstanding deal. The notional backing this IO is the outstanding balance of the deal.[8] As discussed above, the **OC account** is equal to the difference between the outstanding principal on the collateral backing the deal and the outstanding principal on the bonds. On average, the OC account consists of 0.35% to 0.55% of the deal and is either created at the outset of the deal, or built up over time. As the arrow between the Excess Spread and Over-collateralization accounts in Figure 5 suggests, the build up of the OC account is fueled through Excess Spread.[9]

Figure 5: Credit Enhancement in an OC Structure through Excess Spread and Over-Collateralization

- **Excess Spread or Margin**
  - The difference between the WAC on the collateral and the WAC allocated to the bonds in the structure.
- **Over-collateralization**
  - The difference between the outstanding balance of the collateral and the outstanding balance of the bond.



Source: Banc of America Securities

As the reader has probably sensed at this point, the details of how the OC Structure provides credit enhancement is a little more complicated than in the senior-subordinate shifting interest structure. Here, the role of the first loss piece is not played by the junior most subordinate tranche but instead assumed by excess interest. If excess interest is not sufficient to cover losses in any period, then OC is used to cover the remaining losses. On the other hand, if there are no losses and delinquencies, excess spread is used to build up OC. If the OC is already at its "target" level, excess spread is released to the residual. Once all the bonds are paid off, the outstanding OC balance is also released to the residual.

[8] Note that this balance does not include the OC amount.
[9] It is helpful to walk through an example to understand the concept of excess interest. Let's suppose that the collateral balance backing a deal is $100, the principal balance of the bonds is $99.50, OC is $0.50, the collateral WAC is 6% and the deal WAC is 5%. Then, the initial amount of excess spread is (6% - 5%) * $99.50 plus the initial amount of OC interest (6% * $0.50).

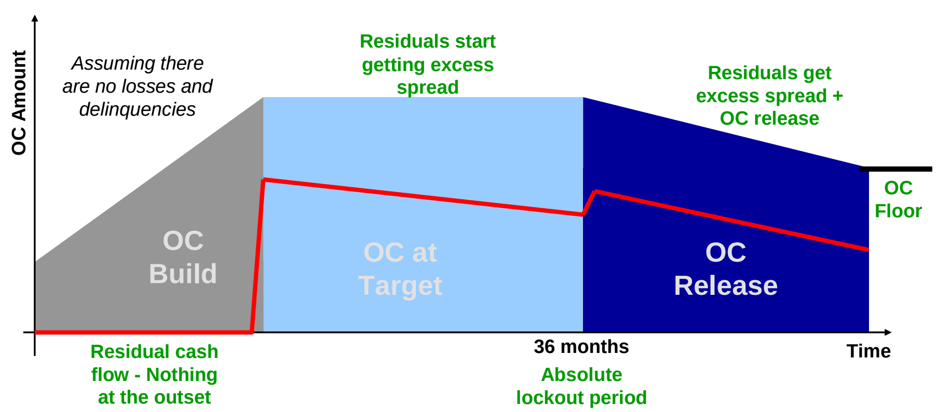
To visualize how all of this works, take a look at Figure 6. The figure provides a visual summary of the relationship between excess spread and OC under the assumption that there are no delinquencies or losses. The "Build", "Target", and "Stepdown" phases highlighted in the figure are basically determined by the **OC requirements**. These requirements consist of a set of rules that define the OC level to be maintained or achieved over different periods in the lifetime of a deal. For example:

- An initial OC target of 0.5% of collateral balance during the first 36 months.
- After month 36, a target of 2 x Initial OC Target x Current Balance, subject to a floor of 0.4% of the original collateral balance.[10]

Finally, the figure also shows that the residual is basically a "mop-up" tranche that soaks up all the remaining cash flows from the excess spread and the OC amount based on fulfilling certain requirements.

Figure 6: The Build Up of Over-Collateralization through Excess Spread

- The allocation of excess spread toward OC is typically structured in three phases:
  - **OC Build**
    - Excess spread is used to build up OC to its target level. No excess spread or OC is transferred to the residual.
  - **OC at Target**
    - The OC level has been built up to the targeted level. The residual gets the excess spread.
  - **OC Stepdown**
    - Once the stepdown is triggered, the OC level is typically allowed to decrease in proportion to the current balance. The residual then gets the excess spread and the OC in excess of the target up to the OC floor.



Source: Banc of America Securities

Now that we have some perspective on how excess interest and OC are used to provide credit enhancement, we take a look at some of the cash flow patterns on the senior, subordinate and residual tranches in an OC structure. For the senior and subordinate tranches in the structure, the situation is similar in principle to what we saw in the senior/sub shifting interest structure in that we have the concept of lockout, stepdown, and triggers. The lockout period is always an absolute three years in an OC structure,

[10] This floor is known as the **OC floor**.

regardless of the collateral type, unlike the case with senior/sub shifting interest structures. After the initial lockout period, during which subordinates are barred from receiving both scheduled *and* unscheduled principal payments, a stepdown may take place provided that OC requirements are met and triggers are passed. If a stepdown occurs, subordinate classes start receiving principal paydowns such that all bonds maintain a specified multiple (typically 2x) of the original credit enhancement based on the current balance. On any payment date, the distribution of cash flows to senior and subordinate tranches is arranged to be such that the maximum proportion of bonds achieve or restore this specified multiple of the original credit enhancement (based on the current balance). The distribution of cash flows that achieves this target is referred to as the **optimal distribution**.

As before, triggers protect the senior tranches by preventing or suspending the stepdown if tripped. A typical set of trigger conditions in an OC structure are:

- **Delinquency Test**: Average 60-day delinquencies on the collateral (over a 2-3 month period) cannot exceed 50% of the senior credit enhancement percentage. For example, if the senior credit enhancement percentage is 14% based on the optimal distribution target, then average 60-day delinquencies cannot exceed 7% of the current collateral balance for the trigger to pass.
- **Cumulative Loss Test**: Cumulative losses must be less than a specified schedule. A typical schedule is given in Figure 7.

Figure 7. Trigger Condition for Cumulative Loss Test in a Typical OC Structure

Month <=	Max. Cum. Loss as % of Orig. Collateral Balance
36	0.6
48	1.1
60	1.6
72	2.2
>72	2.4

Source: Banc of America Securities

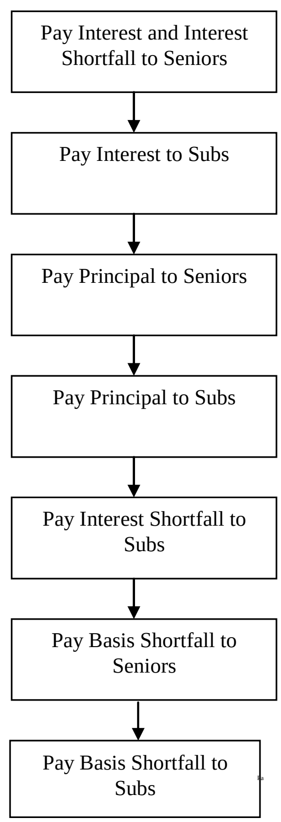
Another nuance to understanding OC (and shifting interest) structures is the concept of **shortfall**. **Interest shortfall** refers to the situation in which a deal becomes **under-collateralized**. This rare event occurs if a senior tranche is barred from write-downs while the underlying collateral experiences heavy losses. In this situation, accrued interest on the collateral is less than the accrued interest due on the bonds. The **interest shortfall** is covered through excess interest. While excess interest provides a mechanism for recovering interest shortfalls in OC structures, no such protection is available in a shifting interest structure.

Basis shortfall (also called **coupon cap shortfall**) arises when floating rate bonds are created in a structure. The shortfall is equal to the difference between the accrual rate on the bonds issued by the structure (@LIBOR + margin) and the **coupon cap rate**. The coupon cap rate is the net accrual rate on the collateral converted to the bond day count basis. The conversion is necessary because interest is calculated on mortgages on a 30/360 basis while LIBOR floaters have an Actual/360 day basis. In scenarios where a basis shortfall occurs, the interest amount based on the coupon cap rate has a higher payment priority than the basis shortfall. Basis shortfall may occur over multiple periods and is collected in a separate account which may or may not be interest bearing. The residual only receives its quota of excess interest once the basis shortfall is covered.

For example, consider a 1-month LIBOR floater in a deal backed by fixed-rate collateral with a coupon cap rate of 6.50%. If the current value of 1-month LIBOR is 5.30% and the weighted-average margin on the LIBOR floaters is 50bps, then will the deal will have excess interest of 70bps. Now, if 1-month LIBOR increases by 100bps, a basis shortfall of 30bps will occur in the next period. If rates stay at this level for another month, another 30bps of basis shortfall will be added to the outstanding amount from the previous period.

Figure 8 summarizes a typical **waterfall** (flow of funds) in an OC structure.

Figure 8. Flow of Funds in a Typical OC Structure



Source: Banc of America Securities

III. COMPARING SHIFTING INTEREST AND OC STRUCTURES

Armed with a general sense for how shifting interest and OC structures work, we now look to obtain deeper insight by working through a detailed analysis of the cash flows associated with the various tranches of these structures. Figure 9 summarizes the deal structure of two hypothetical deals structured to shifting interest and OC conventions.[11] To facilitate an apples-to-apples comparison, both deals are backed by identical fixed-rate collateral with a weighted-average coupon (WAC) of 7.40%.[12] Both the deals support fixed-rate bonds with a pass-through rate of 5.90%. The excess interest is used to create a WAC IO in the shifting interest deal, while it provides additional subordination in the OC structure.

Figure 9: Initial Deal Structure for Shifting Interest and OC Structures Backed By Fixed-Rate Collateral

Senior/Sub Shifting Interest Structure

Tranche	Estimated Rating	Balance	% of Total Collateral	Credit Support (%)
SNR	AAA	917,500,000	91.75%	8.25%
B1	AA	29,500,000	2.95%	5.30%
B2	A	16,000,000	1.60%	3.70%
B3	BBB	10,000,000	1.00%	2.70%
B4	BB	11,000,000	1.10%	1.60%
B5	B	8,500,000	0.85%	0.75%
B6	NR	7,500,000	0.75%	0.00%
Total Bond Size		1,000,000,000	100.00%	

OC Structure

Tranche	Estimated Rating	Balance	% of Total Collateral	Credit Support (%)
SNR	AAA	944,500,000	94.45%	5.55%
M1	AA+	13,000,000	1.30%	4.25%
M2	AA	5,000,000	0.50%	3.75%
M3	AA-	5,000,000	0.50%	3.25%
M4	A+	5,000,000	0.50%	2.75%
M5	A	5,000,000	0.50%	2.25%
M6	A-	5,000,000	0.50%	1.75%
M7	BBB	5,000,000	0.50%	1.25%
M8	BBB-	5,000,000	0.50%	0.75%
M9	BB+	3,500,000	0.35%	0.40%
OC		40 bps		
Total Bond Size		996,000,000	99.60%	

Source: Banc of America Securities

Cash Flow Differences Between a Senior/Sub Shifting Interest and OC Structure Assuming Zero Losses

Figure 10 shows principal payments to the senior and subordinate classes for the two structures assuming a constant prepayment speed of 25% CPR and no defaults. For the sake of clarity, we have aggregated the payments to various subordinate classes into one group and refer to it as "SUB" in Figure 10 and throughout the rest of the primer. Similarly, the senior class is designated as "SNR."

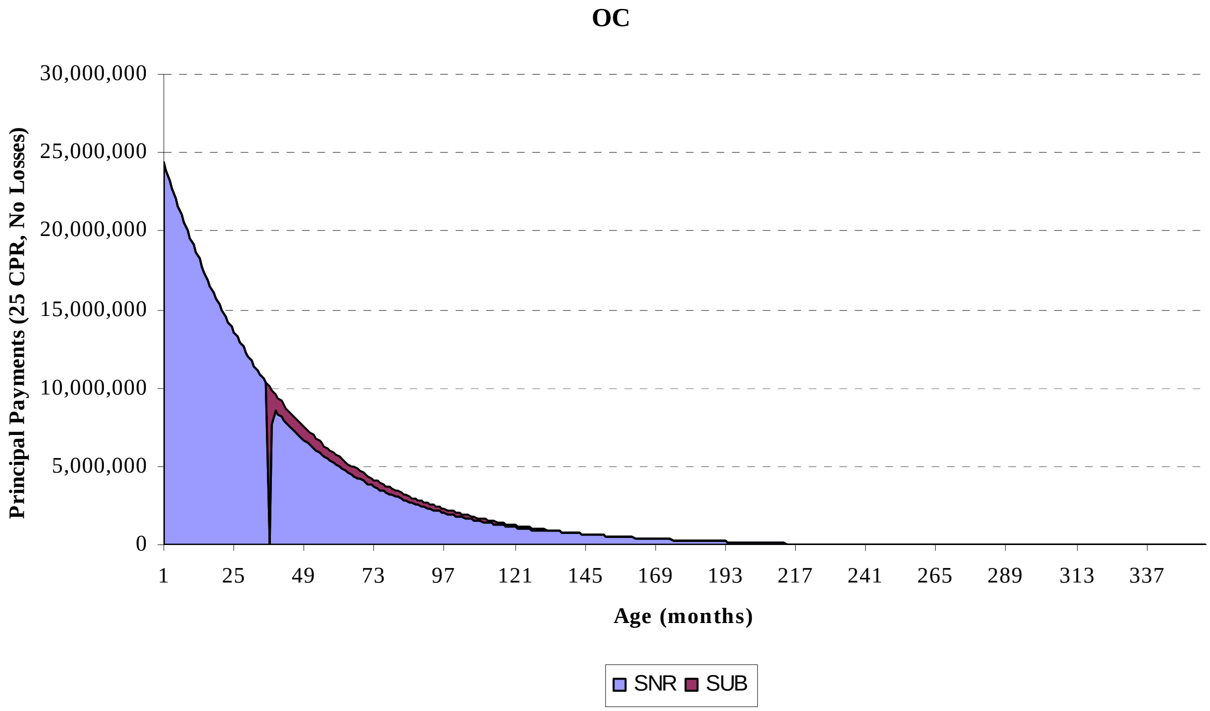
The first point to note is that the subordinates corresponding to the two different structures start getting paid at different points in time due to differences in lockout requirements. In the case of the shifting interest structure, the subordinates are eligible to receive scheduled principal payments on a pro-rata basis from day one, but they start receiving unscheduled principal payments (prepayments) only after the expiration of the lockout period (5 years).

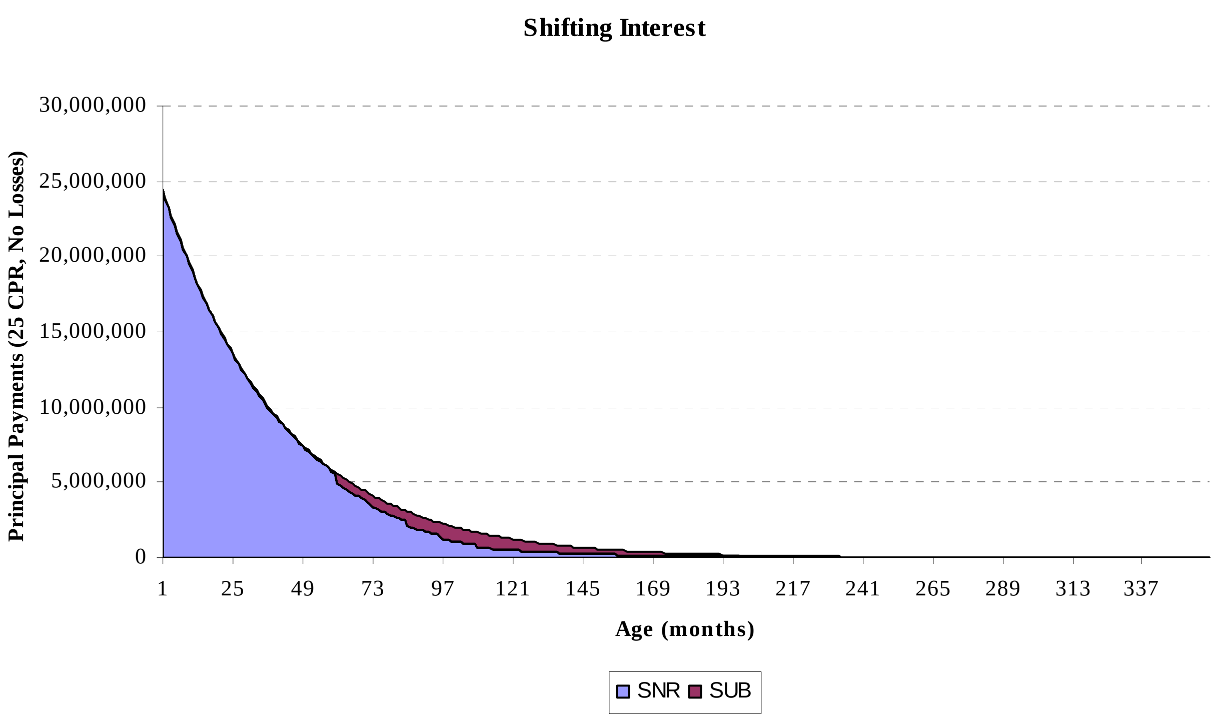
[11] We do not consider the cash flows associated with the residual class in our analysis since there is no counterpart on the shifting interest side.

[12] The collateral characteristics and payment distribution rules for the two deals are summarized in Appendix B.

On the other hand, subordinate bonds in the OC structure start receiving principal payments (both scheduled and unscheduled) only after the expiration of lockout period and when the OC size as a fraction of the current balance becomes twice the initial OC target (40 bps in this case). At 25% CPR, the OC target is met at around month 28. However, the subordinates in the OC structure do not start receiving principal payments until month 37 due to the 3-year absolute lockout. Furthermore, because of the 36 month lockout period in the OC structure, the senior tranche receives all the prepayments during this period. Since the subordinate tranches are totally locked out over this span of time, subordination levels shoot up higher than target levels at the end of 36 months resulting in a sudden release of principal to the subordinates to restore target credit enhancement. The subordinates in the OC structure also get completely paid down earlier than the subordinates in the shifting interest structure due to the OC floor as the OC class is able to provide the required credit enhancement.

Figure 10: Principal Payments at 25% CPR for Senior and Subordinate Bonds for an OC and Shifting Interest Structure





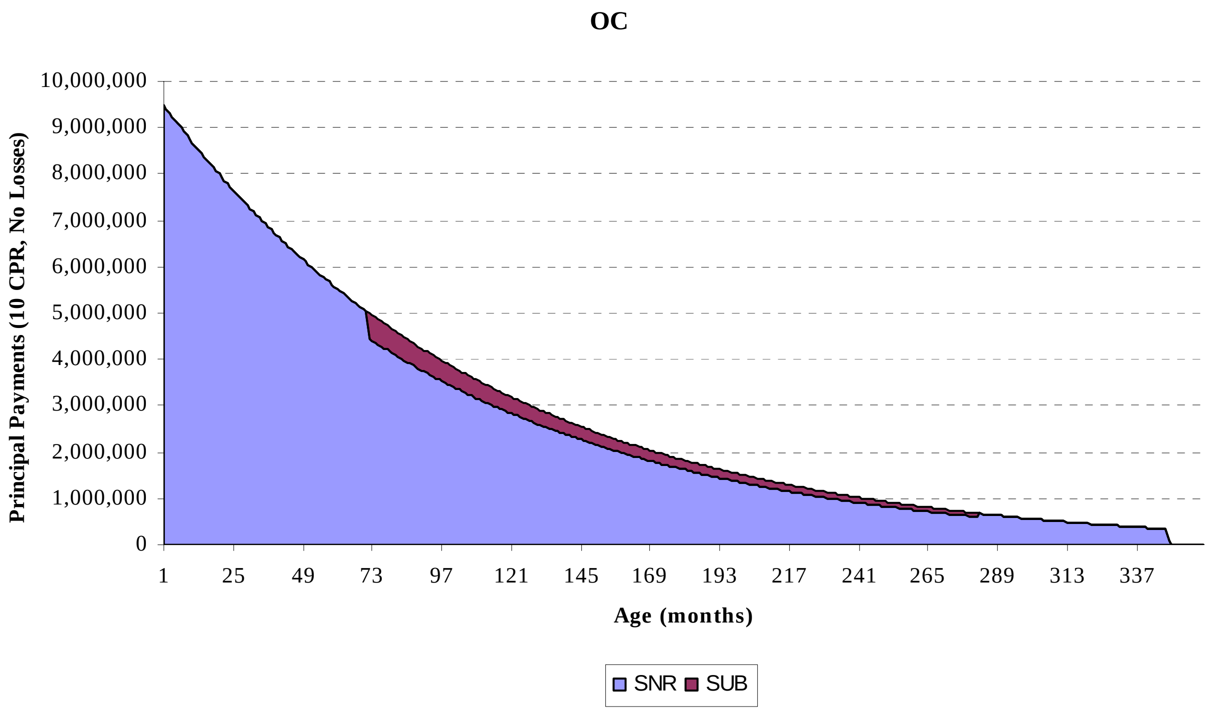
Source: Banc of America Securities

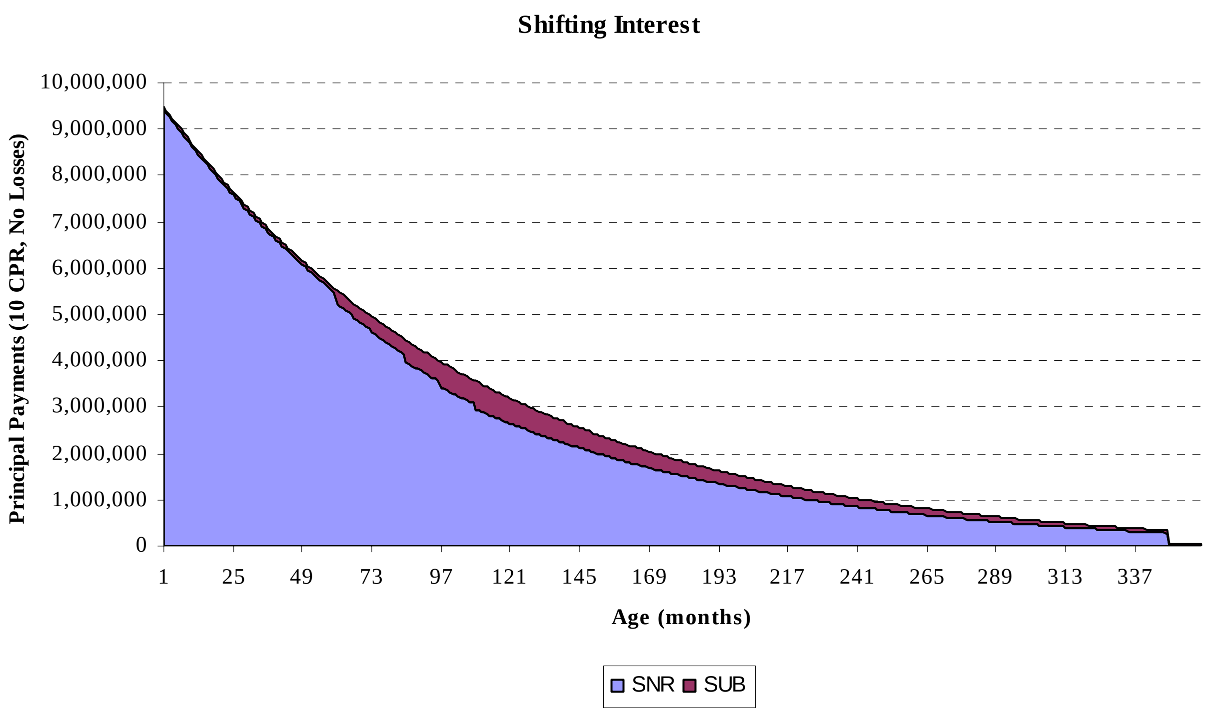
After considering the distribution of cash flows under the two structures in a baseline scenario, we next look at how these structures behave under extreme prepayment scenarios. As in the previous example, we assume that there are no defaults. First, let's look at the case of slow prepayments (10% CPR). The top panel in Figure 11 shows the differences in principal distribution for the two structures. The slower prepayments extend the effective lockout period for the subordinates in the OC structure as the target subordination level is not met. The subordinates start receiving principal cash flows only after month 70. Despite a late start, all the subordinates in the OC structure get paid off earlier than the subordinates in the shifting interest structure due to the OC floor.

Now, consider principal payments for the two structures at fast prepayment speeds of 40% CPR as shown in Figure 12. When the collateral was prepaying at 25% CPR, we saw that the OC structure experiences a sudden release of principal at the end of the 3-year lock out period. Similarly, at 40% CPR, the subordination level sees an even more dramatic increase and hence the principal release takes place for an extended period (four months as opposed to one month in the 25% CPR case). The target credit enhancement is achieved again over a four month period and the senior tranche starts receiving principal payments again.

In the case of the senior/sub shifting interest structure, the senior tranche gets paid off just before the end of the lockout period and before the senior tranche in the OC structure gets paid off. On the other hand, the subordinate class in the OC structure pays off before the subordinates in the shifting interest structure due to the OC floor.

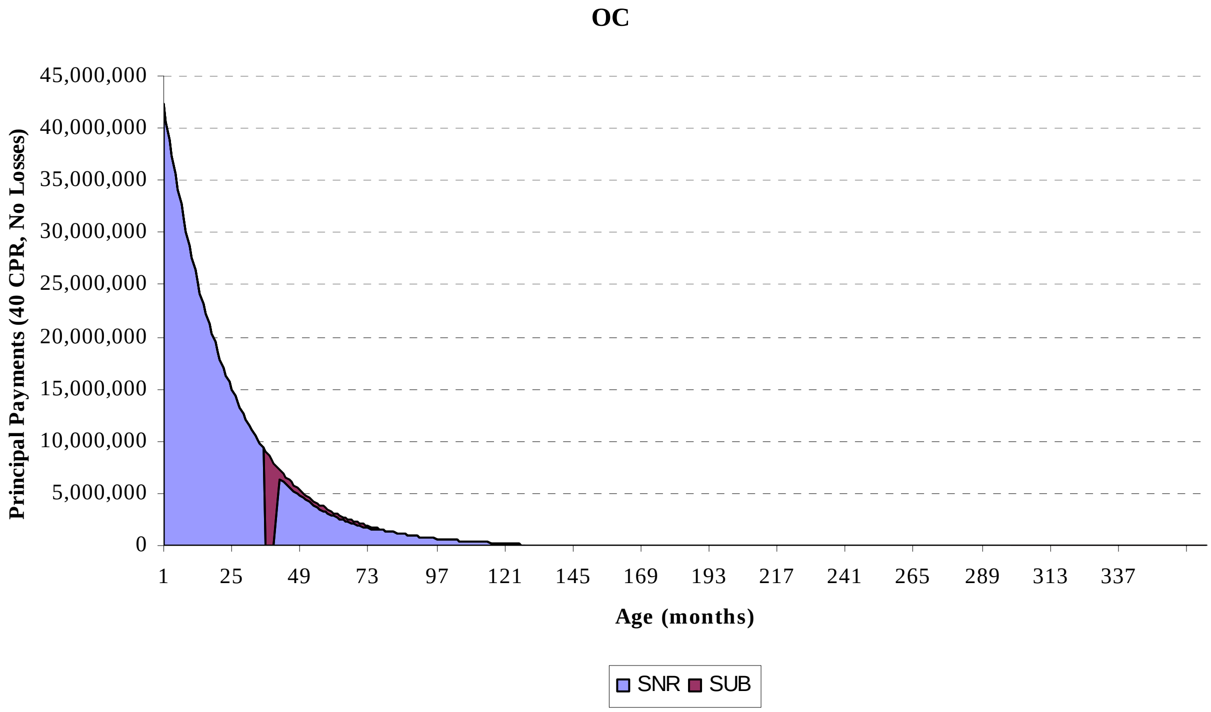
Figure 11: Principal Payments at 10% CPR for Senior and Subordinate Bonds for an OC and Shifting Interest Structure

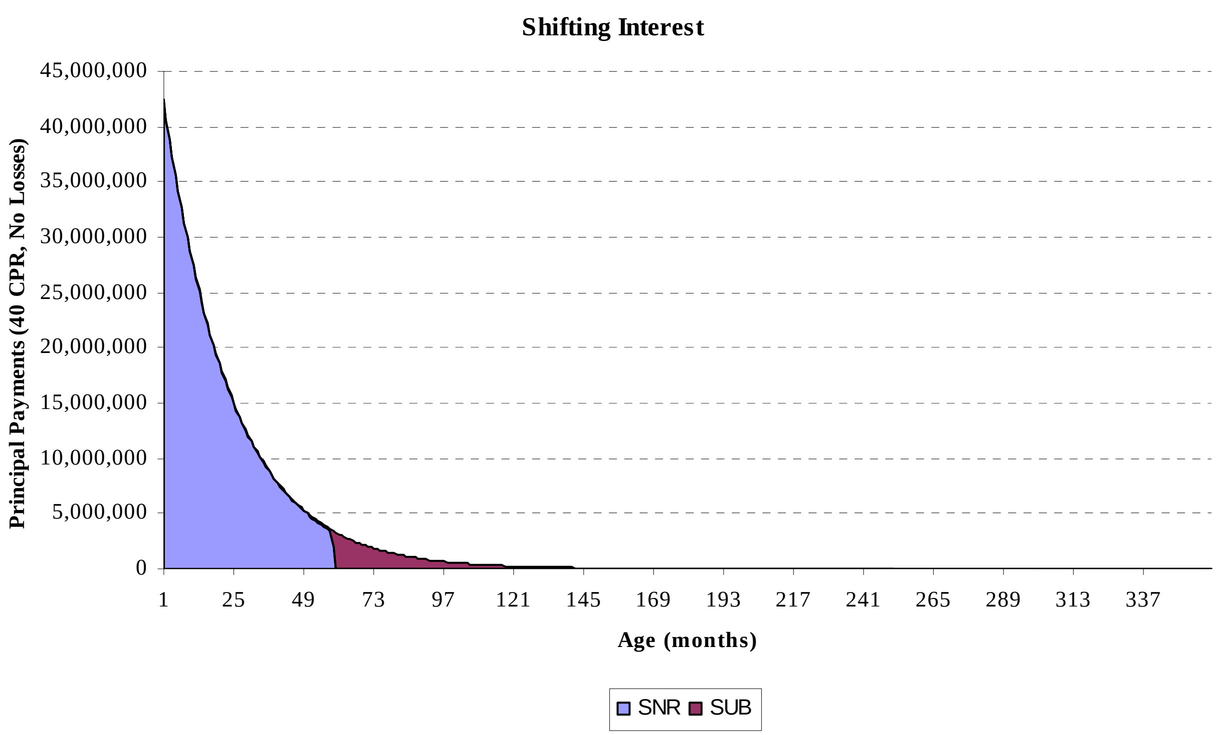




Source: Banc of America Securities

Figure 12: Principal Payments at 40% CPR for Senior and Subordinate bonds for an OC and Shifting Interest Structure





Source: Banc of America Securities

Another perspective on the cash flow distribution patterns in the two structures can be gained by looking at how the weighted-average life (WAL) of comparable tranches changes in different prepayment scenarios. Figure 13 tabulates the WALs of the AAA, AA, A and BBB rated tranches in both structures for the three prepayment scenarios considered above in Figures 10-12. Except for the most senior tranche, the WALs for all other tranches are shorter in the case of the OC structure. The requirement of an OC floor in the over-collateralization structure helps maintain the target subordination and allows the subordinates to be paid down earlier as a result of optimal distribution. Among the different subordinate classes, the lowest rated bonds get paid down earlier resulting in shorter WALs as we move down in credit. In the case of the shifting interest structure, the subordinates are paid off on a pro-rata basis resulting in longer WALs. The pro rata distribution of principal also results in the same WAL for all the credit tranches in a particular prepay scenario for the shifting interest structure. Of course, as a result of the distribution logic which shortens the WAL of the corresponding subordinates in the OC structure, the WAL of the senior tranche in the OC structure is slightly longer than its counterpart on the shifting interest side.

The reduction in the WALs of subordinates in the OC structure relative to the subordinates in the shifting interest structure is more pronounced under faster prepayment speeds. This is a result of the longer lockout period in the case of the shifting interest structure. At slower speeds, since the subordinates in the OC structure can also extend beyond 5-years, the difference between the WALs of corresponding subordinates in the two structures is not that significant. However, as prepayment speeds increase, the effect of the longer lockout period in the shifting interest structure begins to make its presence felt.

Wrapping up, the subordinates in an OC structure have shorter WALs and worse negative convexity than similarly rated subordinates in a shifting interest structure. Although the results we obtained above were under a zero-default assumption, they continue to hold under more realistic default assumptions.

Figure 13: WAL as a Function of Prepayment Speed for Different Tranches in OC and Shifting Interest Structures

	10% CPR		**25% CPR**		**40% CPR**	
	Shift. Int.	**OC**	**Shift. Int.**	**OC**	**Shift. Int.**	**OC**
AAA	7.43	7.57	2.85	3.23	1.56	1.85
AA	12.98	12.73	9.40	5.45	6.62	3.82
A	12.98	12.39	9.40	5.25	6.62	3.63
BBB	12.98	11.69	9.40	4.90	6.62	3.40

Source: Banc of America Securities

Cash Flow Differences Between a Senior/Sub Shifting Interest and OC Structure Assuming Non-Zero Losses

Having considered the impact of prepayments on cash flows for various tranches in the two structures, we now explore how the two structures behave when the underlying collateral also incurs losses. For the purposes of this analysis, we compare losses on similarly rated tranches in the OC and shifting interest structures under various prepayment and loss scenarios. The results are presented in Figure 14.[13] The loss

[13] Note that some of the high loss scenarios considered in this analysis are extremely unlikely and have been considered for discussion purposes only.

scenarios considered assume that rates evolve along the forward curve, loss severity is 35%, and the recovery lag is 12 months. The table at the bottom of Figure 14 shows the total losses as a percentage of original collateral balance under the different scenarios considered. For example, at 1% CDR and 20% CPR, losses add up to 1.4% of the original collateral balance. Note how the cumulative losses on the collateral decrease steadily under all the default scenarios as prepayment speeds increase. This is because the losses on each factor date are computed as a constant percentage (CDR) of the outstanding balance, which is lower at higher prepayment speeds.

Due to its reliance on excess interest (an IO-like cash flow) for providing additional credit enhancement, the OC structure provides better protection in slow prepayment scenarios across both the senior and subordinate tranches. Comparing the senior tranches in the two structures first, notice that in both structures these bonds incur losses only when extremely high default rates are coupled with slow prepayment speeds. For the OC structure, the senior tranche does not incur any loss up to 5% CDR under all prepayment scenarios. The senior tranche in the shifting interest structure take a hit at 5% CDR when prepayment speeds are less than or equal to 20% CPR.

For the subordinate classes in both structures, the enhanced protection offered by the OC structure at slower prepayment speeds relative to the shifting interest structure is even more pronounced. For example, for the AA tranche in the OC structure, excess interest and the OC are sufficient to absorb all losses up to 3% CDR in all the prepayment scenarios. In contrast, in a shifting interest structure, the AA tranche loses 63% of its original balance at 3% CDR when the prepayment speed is 10% CPR.

Summary

The major differences between typical shifting interest and OC structures are as follows:

- Subordinate classes in the two structures start getting paid at different times due to differences in lockout requirements. In the shifting interest structure, subs are eligible to receive scheduled principal payments on a pro rata basis from day one. They start receiving unscheduled principal payments only after the expiration of the lockout period (5 years). On the other hand, subordinate bonds in an OC structure start receiving principal payments (both scheduled and unscheduled) only after the expiration of lockout period, and only when the OC amount as a fraction of the current balance becomes twice the initial OC target.
- The senior tranche in an OC structure will generally have a slightly longer WAL than the similarly structured senior tranche in the shifting interest structure.
- In a shifting interest structure, subordinates are paid off on a pro rata basis which results in the same WALs for all the credit tranches in any prepayment scenario. In OC structures, different subordinate tranches will have different WALs.
- In most realistic loss scenarios, subordinates in an OC structure will have shorter WALs and worse negative convexity than similarly rated subordinates in a shifting interest structure.
- Due to a reliance on excess interest (an IO-like cash flow) for providing additional credit enhancement, the OC structure provides better protection in slow prepayment scenarios across both senior and subordinate tranches.

Figure 14: Losses on Senior and Subordinate Bonds as a Percentage of their Original Balance for OC and Shifting Interest Structures

OC

Senior

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	0.0%	0.0%	0.0%	0.0%	0.0%
	3	0.0%	0.0%	0.0%	0.0%	0.0%
	5	0.0%	0.0%	0.0%	0.0%	0.0%
	7	1.9%	0.0%	0.0%	0.0%	0.0%
	10	6.2%	2.5%	0.4%	0.0%	0.0%

AA M2

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	0.0%	0.0%	0.0%	0.0%	0.0%
	3	0.0%	0.0%	0.0%	0.0%	0.0%
	5	37.5%	0.0%	0.0%	0.0%	0.0%
	7	100.0%	100.0%	11.2%	0.0%	0.0%
	10	100.0%	100.0%	100.0%	100.0%	17.7%

BBB M7

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	0.0%	0.0%	0.0%	0.0%	0.0%
	3	0.0%	0.0%	0.0%	0.0%	0.0%
	5	100.0%	100.0%	100.0%	100.0%	64.8%
	7	100.0%	100.0%	100.0%	100.0%	100.0%
	10	100.0%	100.0%	100.0%	100.0%	100.0%

Collat

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	1.3%	0.7%	0.5%	0.3%	0.3%
	1	2.6%	1.4%	0.9%	0.7%	0.5%
	3	6.9%	4.0%	2.7%	2.0%	1.5%
	5	10.3%	6.3%	4.3%	3.2%	2.4%
	7	13.1%	8.3%	5.8%	4.4%	3.4%
	10	16.4%	10.9%	7.9%	6.0%	4.7%

Shifting Interest

Senior

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	0.0%	0.0%	0.0%	0.0%	0.0%
	3	0.0%	0.0%	0.0%	0.0%	0.0%
	5	2.7%	0.0%	0.0%	0.0%	0.0%
	7	5.6%	0.4%	0.0%	0.0%	0.0%
	10	9.1%	3.1%	0.0%	0.0%	0.0%

AA B1

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	0.0%	0.0%	0.0%	0.0%	0.0%
	3	63.2%	0.3%	0.0%	0.0%	0.0%
	5	92.1%	39.5%	0.0%	0.0%	0.0%
	7	95.0%	92.4%	22.8%	0.1%	0.0%
	10	96.6%	96.0%	90.4%	26.5%	0.6%

BBB B3

		CPR (%) 10	20	30	40	50
CDR (%)	**0.5**	0.0%	0.0%	0.0%	0.0%	0.0%
	1	18.0%	0.0%	0.0%	0.0%	0.0%
	3	94.7%	91.9%	13.2%	0.1%	0.0%
	5	96.9%	96.5%	95.6%	53.5%	4.0%
	7	97.6%	97.4%	97.2%	96.7%	65.8%
	10	98.1%	98.0%	97.9%	97.8%	97.6%

Source: Banc of America Securities

IV. COMPARING SHIFTING INTEREST STRUCTURES BACKED BY FIXED-RATE VERSUS ADJUSTABLE-RATE MORTGAGES

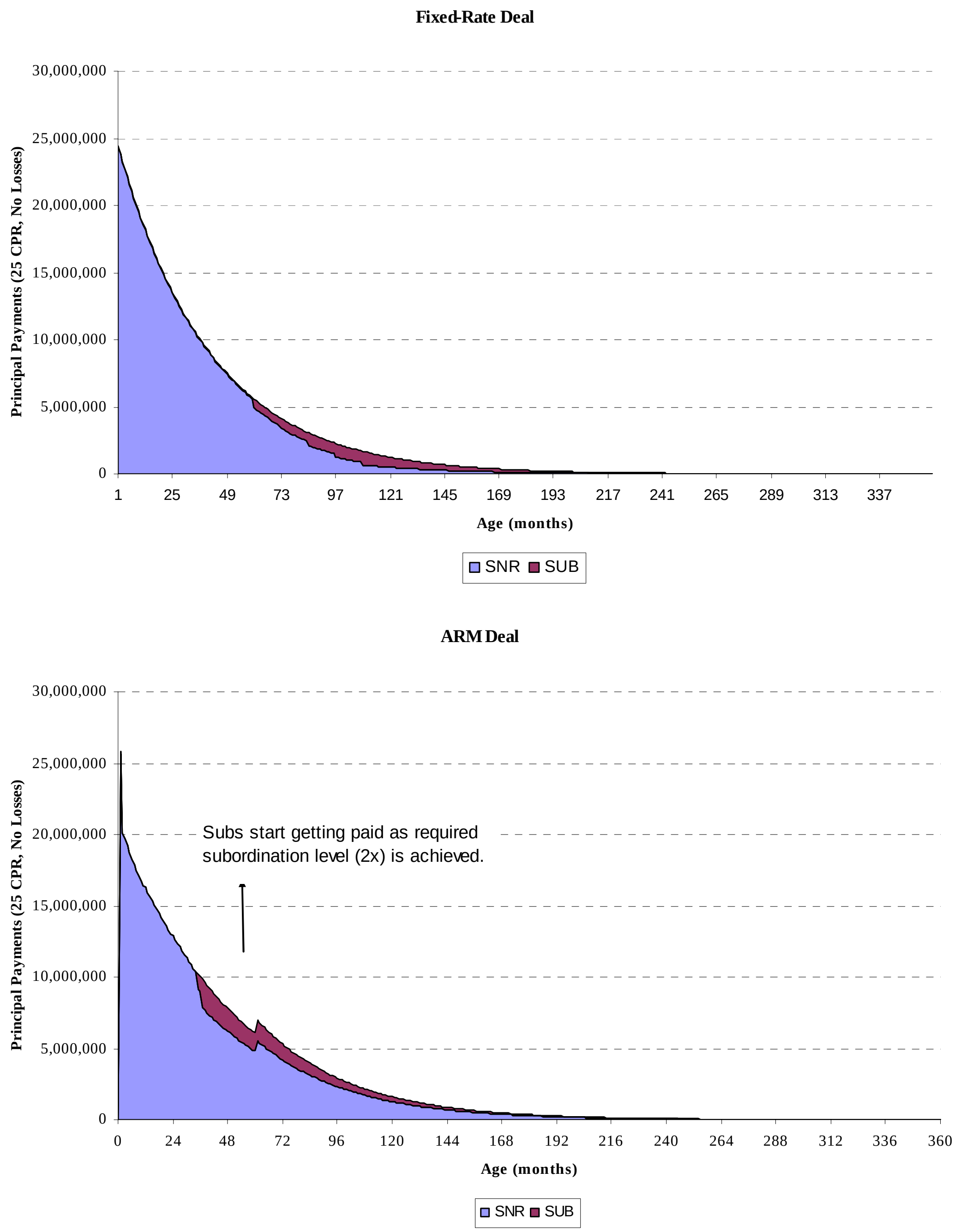
In this section, we consider how the underlying mortgage type can affect the cash flow distribution to various tranches in a structure by considering a shifting interest structure backed by Option ARM collateral. In general, differences between similar tranches of deals backed by fixed-rate and ARM collateral arise because of the following reasons:

- Collateral features such as negative amortization, payment resets and recasts affect the cash flow profiles of different tranches in a deal. This affects both shifting interest and over-collateralization structures.
- The length of the lockout and the nature of lockout (absolute versus conditional) for a typical fixed-rate deal are different from that of an ARM deal in shifting interest structures. There is no difference in the lockout provisions for FRM vs ARM collateral in OC structures.

Conditional Lockout Versus Absolute Lockout

As discussed earlier, most shifting interest structures backed by ARM collateral have a conditional lockout while those backed by fixed-rate collateral have an absolute lockout. In an ARM deal, if subordination levels reach 2x the original subordination during the first 3 years, then subordinates receive 50% of their pro rata share of prepayments. After 3 years, reaching 2x enhanced subordination level results in the subordinates receiving 100% of their pro rata prepayment share. Figure 15 shows principal payments to the senior and subordinate classes assuming a constant prepayment speed of 25% CPR and no defaults in senior/sub structures using fixed-rate and ARM collateral. Note how the conditional lockout enables the SUB in an ARM deal to receive principal earlier than the SUB in the fixed-rate deal.

Figure 15: Effect of Conditional Lockout Versus Absolute Lockout on Shifting Interest Structures

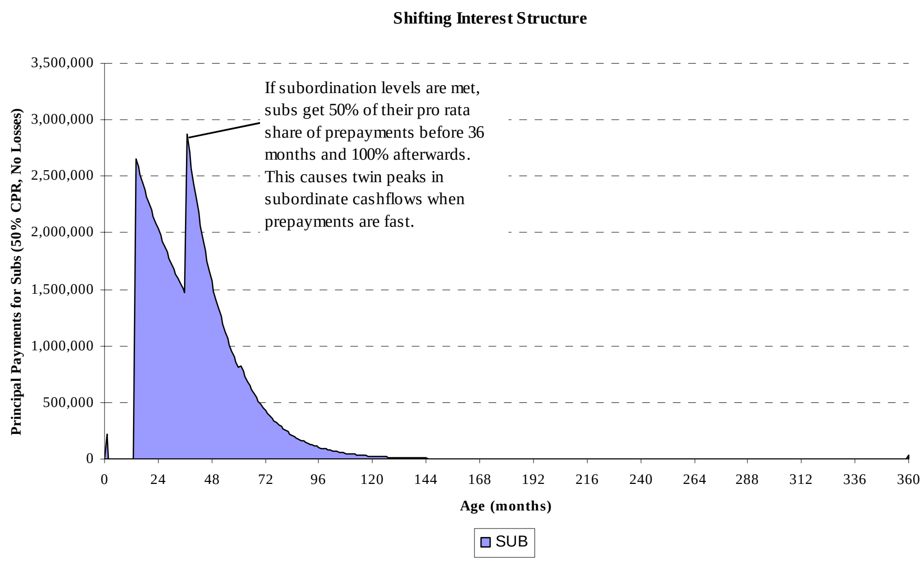


Source: Banc of America Securities

Two Spikes in Principal Payments to Subordinates of ARM Deals at High Speeds

Another difference between the cash flow profiles of SUBs in the two deals arises because of the early distribution rule used in ARM deals. We show cash flow distributions to SUB in a 50% CPR scenario for the shifting interest structure backed by ARM collateral in Figure 16. Unlike a structure backed by fixed-rate collateral, we see two distinct spikes in principal payments. The first spike occurs when the subordination level of the senior class reaches twice the original subordination level. At this point subordinates start receiving 50% of their pro rata share of prepayments. The second spike occurs at month 37, at this point subordinates start getting 100% of their pro rata share of prepayments.

Figure 16: Principal Payments to Subordinate Bonds at 50% CPR for a Shifting Interest Structure



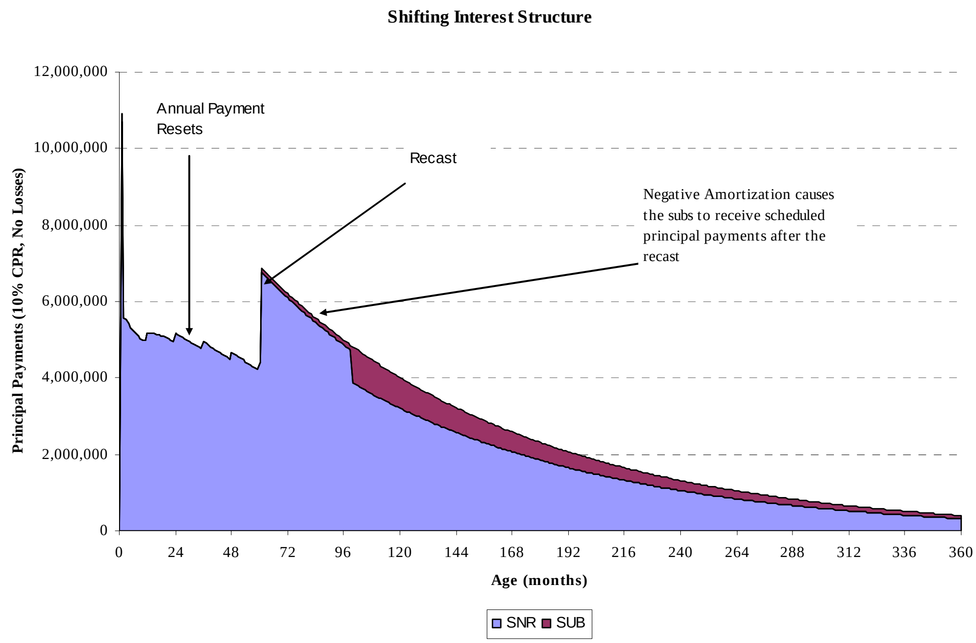
Source: Banc of America Securities

The Impact of ARM Features on Cash Flows

Finally, ARM features such as negative amortization, payment resets and recasts affect cash flow profiles of different tranches in shifting interest and OC structures. In Figure 17 we show the effect of negative amortization, payment resets and recast on the cash flows to senior and SUB tranches in a 10% CPR scenario. We can draw the following conclusions:

– There are small jumps in principal payments during the first four years due to annual payment resets and a major jump at the beginning of the 5^{th} year anniversary due to a payment recast on the collateral.
– In the case of the shifting interest structure, although the subordinates are entitled to receive their pro rata share of scheduled principal payments from the beginning, negative amortization results in these bonds starting to receive their scheduled principal payments only after the recast as enough scheduled principal becomes available.

Figure 17: Principal Payments for Senior and Subordinate bonds in an ARMs Deal



Source: Banc of America Securities

APPENDIX A: LOAN-LEVEL STRUCTURING OF MORTGAGE COLLATERAL

Fixed-Rate Collateral

As we discussed earlier, the tranching shown in Figure 1 represents a simplified version of what happens in reality; both the senior and the subordinate classes of the structure are subject to further structuring depending upon market demand. Figure 18 presents a more realistic example. Note that the starting point for the structuring process is the raw loan package less an IO strip which is used to pay servicing and trustee fees. The end-servicer of the loan package is required to keep at least 25bps of IO, while the trustee fee averages 1bp of IO.

The tranching of the senior class into a fixed-rate pass-through, WAC IO and PO results from the requirement that the coupon on the AAA tranche be fixed for the duration of the deal. The problem is that since virtually all loan packages will have loans with different note rates, and that since loans with different note rates prepay differently, the weighted-average coupon (GWAC) of the raw loan package will "drift" over time. This would mean that the coupon on the AAA package, which is calculated as the difference between the GWAC on the loan package and a fixed coupon IO strip[14] would not be fixed over time. The structuring solution to this problem is to fix the coupon on the AAA tranche by creating two additional tranches, a WAC IO and a WAC PO, which bear the risk of coupon drift over time. This is best understood through an example.

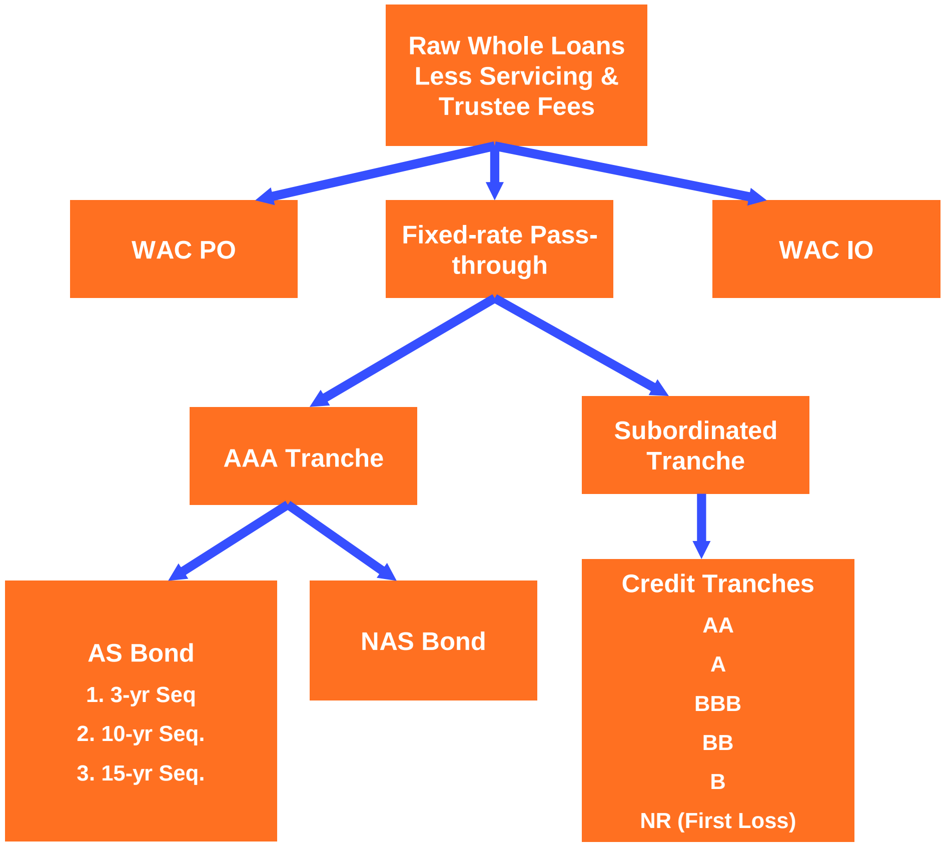
Suppose we have a $300 million pool of loans backing the AAA tranche with a GWAC of 6.01% and a required subordination level of 3%. For simplicity, we will assume that there is no excess servicing involved (the total servicing fees is 25bps) and the trustee fee is 1bp. Assume that after subtracting the servicing and trustee fee IO strip that the net coupons of the pool are evenly distributed between 5.50%, 5.75%, and 6.00%. We can maintain a fixed coupon of 5.75% over the life of the deal as follows. Leave the $100 million loans with a net coupon of 5.75% as is. The $100 million loans with 6.0% net coupons are stripped down to 5.75% by creating a $100mm WAC IO with an initial coupon of 25bps. The $100 million loans with 5.5% net coupons are used to create a WAC PO with a notional value of (1 – 5.50/5.75) * $100 million = $4.35 million. The point is that the interest received on the principal backing the PO can be diverted to gross up the coupon on the remaining $95.65 million pool to 5.75%.

Since the total credit support is 3%, 0.97 * 300 = $291 million is available for the senior balances (AAA, WAC IO and WAC PO) and $9mm for the subordinate tranches (AA, A, BBB, BB, B, NR). The size of the AAA pass-through class is $291 million - $4.35 million = $286.65 million.

As Figure 18 shows, the fixed-rate pass-through can be further tranched into a NAS and AS class. The NAS bond is structured similarly to the subordinate class in terms of the lockout and shifting of prepaid principal. In most cases, the NAS is also locked out of scheduled principal payments. The AS portion is typically structured into three sequential classes with increasing weighted-average lives. Notice that since the Non-Agency AAA fixed-rate pass-through is very similar to an Agency pass-through pool from a cash-flow and credit perspective, we can replicate any structure that could be created in the Agency CMO sector. So, for instance, it would be possible to create a PAC/Support structure from the fixed-rate class.

[14] The servicing and trustee fees.

Figure 18: Tranches in a Senior/Sub Shifting Interest Structure



Source: Banc of America Securities

Adjustable-Rate Collateral

Since the mortgage rate on an ARM is not fixed, we do not require the net coupon on the various tranches in the structure to be fixed for the deal. This is in contrast to the situation for non-agency securities collateralized by a pool of fixed-rate mortgages described earlier. In particular, no WAC IO or PO tranche is created in the process of structuring a pool of non-agency ARMs. However, there is something conceptually similar to a WAC IO tranche called an Excess IO tranche that is created when we structure Short Reset ARMs. A Short Reset ARM is an ARM with an initial period that is a year or less. The AAA tranche structured off of short reset collateral is priced at par, which implies a certain discount margin over the ARM index. Because the typical margin on ARM collateral is between 175bp to 225bp for LIBOR-based ARMS, while the discount margin on the AAA par floater is around 17bp, a tranche is created out of the excess margin on the underlying collateral.

APPENDIX B: PAYMENT RULES FOR FIXED-RATE COLLATERAL

Collateral Characteristics
Gross Coupon: 7.40%
Servicing Fee: 0.25%
Term: 360 months
Prepayment Penalty: None

Senior/Subordinate Shifting Interest Structure

Flow of Funds: Pay interest and principal to all tranches in order of seniority and in accordance with the stepdown/lockout restrictions.

Lockout: Subordinates are entitled to receive scheduled principal payments on a pro rata basis from Day 1. However, unscheduled principal payments are locked out for 5 years. After 5 years, if the triggers are passed, unscheduled principal payments are made to subordinates according to the shifting interest schedule.

Step Down:

Shifting Interest Schedule: If triggers are passed, the pro rata share of prepayments owed to the subordinate classes received by seniors during the lockout period decrease according to the schedule in Figure 19.

Figure 19: Shifting Interest Prepayment Lockout Schedule

5-year Lockout *Month <=*	**Shift %**
60	100
72	70
84	60
96	40
108	20
120	0

Source: Banc of America Securities

Early Release of Principal: Early release of principal to subordinates before the end of the lockout period is only possible if the senior bonds are completely paid down.

Triggers:

- 6-month Average 60+ Day Delinquency Balance < 50% of Current Subordinate Balance
- Losses do not exceed specified percentage of initial Subordinate Balance (30% in the first 6 years; 35% at the end of 7 years; 40% at the end of 8 years; 45% at the end of 9 years; 50% at the end of 10 years).

OC Structure

Flow of Funds: Pay interest and principal to all tranches in order of seniority and in accordance with the stepdown/lockout restrictions.

Lockout: Lockout period of 3 years, during which the subordinates are barred from receiving both scheduled and unscheduled principal flows. After the end of 3 years, a step down may take place provided that OC requirements are met and triggers are passed.

Early Release of Principal: Early release of principal to subordinates before the end of the lockout period is only possible if the senior bonds are completely paid down.

OC Requirements:

- Initial OC target of 0.4% of the collateral balance during the first 36 months.
- After month 36, an OC target of 2 x The Initial OC Target x The Current Balance subject to a floor of 0.4% of the original collateral balance.

Triggers:

- The 3-month average of 60-day delinquencies as a percentage of current collateral balance cannot exceed 53% of the required subordination level for the senior class.
- Cumulative losses must be less than the schedule specified in Figure 20.

Figure 20. Trigger Condition for Cumulative Loss Test

Month <=	Max. Cum. Loss as % of Orig. Collateral Balance
36	0.6
48	1.1
60	1.6
72	2.2
>72	2.4

Source: Banc of America Securities

IMPORTANT INFORMATION CONCERNING U.S. TRADING STRATEGISTS

Trading desk material is NOT a research report under U.S. law and is NOT a product of a fixed income research department of Banc of America Securities LLC, Bank of America, N.A. or any of their affiliates (collectively, "BofA"). Analysis and materials prepared by a trading desk are intended for Qualified Institutional Buyers under Rule 144A of the Securities Act of 1933 or equivalent sophisticated investors and market professionals only. Such analyses and materials are being provided to you without regard to your particular circumstances, and any decision to purchase or sell a security is made by you independently without reliance on us.

Any analysis or material that is produced by a trading desk has been prepared by a member of the trading desk who supports underwriting, sales and trading activities.

Trading desk material is provided for information purposes only and is not an offer or a solicitation for the purchase or sale of any financial instrument. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Although information has been obtained from and is based on sources believed to be reliable, we do not guarantee its accuracy, and it may be incomplete or condensed. All opinions, projections and estimates constitute the judgment of the person providing the information as of the date communicated by such person and are subject to change without notice. Prices also are subject to change without notice.

With the exception of disclosure information regarding BofA, materials prepared by its trading desk analysts are based on publicly available information. Facts and ideas in trading desk materials have not been reviewed by and may not reflect information known to professionals in other business areas of BofA, including investment banking personnel.

Neither BofA nor any officer or employee of BofA accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this report or its contents.

To our U.K. clients: trading desk material has been produced by and for the primary benefit of a BofA trading desk. As such, we do not hold out any such research (as defined by U.K. law) as being impartial in relation to the activities of this trading desk.

IMPORTANT CONFLICTS DISCLOSURES

Investors should be aware that BofA engages or may engage in the following activities, which present conflicts of interest:

- The person distributing trading desk material may have previously provided any ideas and strategies discussed in it to BofA's traders, who may already have acted on them.
- BofA does and seeks to do business with the companies referred to in trading desk materials. BofA and its officers, directors, partners and employees, including persons involved in the preparation or issuance of this report (subject to company policy), may from time to time maintain a long or short position in, or purchase or sell a position in, hold or act as market-makers or advisors, brokers or commercial and/or investment bankers in relation to the products discussed in trading desk materials or in securities (or related securities, financial products, options, warrants, rights or derivatives), of companies mentioned in trading desk materials or be represented on the board of such companies. For securities or products recommended by a member of a trading desk in which BofA is not a market maker, BofA usually provides bids and offers and may act as principal in connection with transactions involving such securities or products. BofA may engage in these transactions in a manner that is inconsistent with or contrary to any recommendations made in trading desk material.
- Members of a trading desk are compensated based on, among other things, the profitability of BofA's underwriting, sales and trading activity in securities or products of the relevant asset class, its fixed income department and its overall profitability.
- The person who prepares trading desk material and his or her household members are not permitted to own the securities, products or financial instruments mentioned.
- BofA, through different trading desks or its fixed income research department, may have issued, and may in the future issue, other reports that are inconsistent with, and reach different conclusions from the information presented. Those reports reflect the different assumptions, views and analytical methods of the persons who prepared them and BofA is under no obligation to bring them to the attention of recipients of this communication.

This report is distributed in the U.S. by Banc of America Securities LLC, member NYSE, NASD and SIPC. This report is distributed in Europe by Banc of America Securities Limited, a wholly owned subsidiary of Bank of America NA. It is a member of the London Stock Exchange and is authorized and regulated by the Financial Services Authority.